



Confide

Team collaboration platform with built-in client-contact protection

Company: Algotix AI

Live application: <https://teams-collaboration-tool-bwfb.vercel.app>

Document: Project documentation, v1.0

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1. What is Confide?

Confide is a chat workspace where an agency's team and its clients talk in one place — and where the platform itself makes sure nobody can quietly exchange personal contact details.

Agencies have a common problem: a team member and a client meet inside a project, swap phone numbers or emails, and take the work outside the agency. Normal chat tools (WhatsApp, Slack) cannot stop this, because they never look at what is being sent.

Confide is built around one idea: **the conversation is open, but contact details are not**. Every message is checked on the server before anyone sees it. If it contains a phone number, an email address, a social-media link, a payment handle, or a crypto address — even written in a disguised way — the message is held back and sent to an admin for review instead of being delivered.

In one sentence: Confide gives teams and clients a modern group chat while protecting the agency's client relationships automatically, and keeps tamper-proof evidence of everything that happens.

The four pillars

PILLAR	WHAT IT MEANS
Screened messages	Every message is scanned server-side before delivery. Contact details are caught and held for admin review — the sender sees a policy notice, the group never sees the message unless it is approved.
Masked identities	The admin decides everyone's display name (for example "Ahmed K."). Members cannot rename themselves into a contact card — name changes go through an admin-reviewed request, and even names are screened for contact details.
Admin control	Admins invite people, set names and roles, create and archive groups, review held messages, and see workspace-wide statistics on a dashboard only they can open.
Tamper-proof audit	Every action — login, message, file, approval, name change — is written to an append-only log where each entry is cryptographically linked to the previous one. If anyone edits history, verification reports the exact entry where the chain breaks.

Who uses it

ROLE	WHAT THEY CAN DO
Admin	Everything: dashboard, all groups, moderation queue, name-change reviews, audit log, invitations, group lifecycle, own profile edits (still screened).
Team member	Chat and share files in their groups, request a name change, pick a theme. Lands directly in Chat after login.
Client	Same chat experience as members, inside the groups they are invited to. Sees only masked names of the team.

2. Main features

Group chat

- **Real-time messaging** — messages appear for everyone instantly, with no page reload.
- **Read receipts, WhatsApp style** — one grey tick when delivered, double teal tick only when *everyone* in the group has read the message.
- **Online / offline dots** — a Slack-style presence dot on every avatar.
- **File sharing** — up to 50 MB per file, with a live upload percentage. Oversized files are refused *before* uploading starts, with a clear message. Failed uploads can be retried with one click.
- **Desktop notifications** — Slack-style system notifications when a message arrives while the app tab is in the background. Held messages notify nobody until they are approved.
- **Rich composer** — bold, italic, lists, quotes, code, @mentions.

Message screening & moderation

- Five detector families run on the server: **phone numbers, email addresses, social-media links, payment handles, and crypto addresses**.
- Detection normalizes the text first, so spacing and punctuation tricks (like "0 3 2 1 – 1 2 3...") are still caught.
- Caught messages go to the **moderation queue**. Admins see the message exactly as typed with the caught parts highlighted, who sent it, in which group, and how long it has been waiting.
- **Approve** delivers the message to the group. **Block** replaces it with a policy notice for the sender.
- If a held message waits too long (30 minutes by default, adjustable per workspace and per group), it is **escalated** — a background job sweeps every minute and alerts managers.
- Admins see a **live red dot** on the sidebar and bell the moment something needs review — no reload needed. It clears itself when the queue is empty.

Identity & names

- The admin types each person's display name at invitation time; invitees never get a field to change it.
- Members request name changes; admins approve or reject them on a dedicated review page.
- Names and job titles are screened by the same detectors as messages — a name cannot become a phone number.

Audit log

- Every event is recorded: logins, messages, files, approvals, blocks, invites, name changes, group changes.
- Entries form a **hash chain** — each one is cryptographically linked to the one before it. A scheduled job and a "Verify now" button both check the chain and report the first broken entry if anything was altered.
- Filters by group, member, and event type answer questions like "what did this person do in this group?" without touching the database. Full **CSV export** for reports.

Workspace & comfort

- **Email invitations** — sent from `noreply@algotix.ai`, valid for 7 days, with a copyable backup link for the admin.
- **Admin dashboard** — messages held/delivered, people counts, recent activity, chain status. Members and clients land straight in Chat.
- **Six color themes** — chosen at first login, changeable any time, saved to the account so they follow the user to any device.
- **⌘K quick switch** — jump to any page or group from the keyboard.
- **Responsive** — full mobile layout with a slide-out sidebar; the desktop sidebar collapses ChatGPT-style.

3. How message screening works

The journey of one message, from the sender's keyboard to the group.

- 1 The sender presses Send.**
The message goes to the server — never directly to other people.
- 2 The server scans it.**
The text is normalized (spacing, punctuation and disguise tricks removed) and run through all five detector families.
- 3 Clean message → delivered.**
It appears in the group for everyone within about a second, and the audit log records it.
- 4 Contact details found → held.**
The group never sees it. The sender sees it marked as held with a policy notice. Nobody is notified about its content.
- 5 Admins are alerted live.**
A red dot appears on the sidebar and bell within ~2 seconds, plus a toast on the moderation page itself.
- 6 An admin reviews it.**
The queue shows the message as typed, with every caught span highlighted and labeled by rule (for example `phone` , `email`).
- 7 Approve or block.**
Approve delivers it to the group (and only then do notifications fire). Block turns the sender's bubble into a policy notice. Both decisions are audited with the reviewer's identity.
- 8 Nothing waits forever.**
A background job checks every minute; anything held longer than the configured limit (default 30 minutes) is escalated to the group's manager.

Why server-side matters: the check happens where users cannot switch it off. There is no client-side toggle, no browser trick, and no app version that skips the scan — the message simply does not exist for other people until the server decides it is safe.

What the detectors catch

DETECTOR	EXAMPLES OF WHAT IS CAUGHT
Phone numbers	Local and international formats, spaced or punctuated digits, spelled-out digit tricks.
Email addresses	Plain addresses and disguised forms ("name at gmail dot com").
Social links	WhatsApp, Telegram, Instagram, LinkedIn, and similar profile or invite links.
Payment handles	PayPal.me, Wise, and similar pay-me handles.
Crypto addresses	Bitcoin / Ethereum-style wallet addresses.

4. Security

Access control

- **Row-Level Security (RLS)** is enforced by the database itself: every query runs as the logged-in user, and the database refuses rows the user should not see. Even a bug in the app cannot leak another group's messages.
- Held messages are visible *only* to reviewers — the RLS policy literally filters them out of everyone else's message list.
- The admin dashboard, moderation queue, and audit log are server-checked on every request, not just hidden in the menu.

Files

- Files live in a **private storage bucket** — there are no public links. Downloads go through the app, which checks group membership first.
- Uploads are capped at 50 MB and refused before transfer if too large.

Evidence

- The audit log is **append-only**: no update or delete, enforced by the database.
- Each entry carries a hash of the previous entry. Editing any historical row breaks every hash after it, and verification points to the exact first broken entry.
- Verification runs on a schedule and on demand; the result (intact / broken at entry #N) is shown on the audit page.

Accounts

- Authentication is handled by Supabase Auth (email + password, secure session cookies).
- Invitations expire after 7 days and are single-use.
- Every login is itself an audited event.

5. Technology & architecture

LAYER	TECHNOLOGY	NOTES
Web application	Next.js 16 (React 19)	Server-rendered pages with client-side navigation; instant skeletons per page.
Database	Supabase — PostgreSQL 17	28 versioned migrations; RLS on every table.
Real-time	Supabase Realtime	Live messages, presence dots, moderation alerts.
File storage	Supabase Storage	Private bucket, membership-checked downloads.
Background jobs	pg_cron	<code>confide-escalations</code> (every minute) and <code>confide-chain-verify</code> (audit chain check).
Email	Resend	Invitations from <code>noreply@algotix.ai</code> (verified domain).
Hosting	Vercel	App served from Tokyo (hnd1), same region as the database for fast queries.
Styling	Tailwind CSS v4	Token-based theme system — 6 themes from one set of CSS variables.
Animations	GSAP + CSS	All animations respect the user's reduced-motion setting.

How the pieces talk

- 1

Browser → Vercel.
The Next.js app serves pages and hosts all API routes (messages, moderation, audit, invitations, files, and so on).
- 2

Vercel → Supabase.
API routes talk to PostgreSQL as the logged-in user, so RLS applies everywhere. Message screening runs here, on the server.
- 3

Supabase → Browser.
Realtime channels push new messages, presence, and moderation alerts straight to open tabs.
- 4

Inside the database.
`pg_cron` runs the escalation sweep and chain verification without any external scheduler.
- 5

Resend.
Called by the invitation route to deliver branded invite emails.

Quality process

- **81 unit tests** cover the detection engine (including a disguise-tricks corpus), scoring, and normalization.
- **End-to-end tests** (Playwright) walk real browser flows: login, invite, chat, hold, approve, audit.
- **CI on GitHub Actions** runs lint, types, unit and e2e tests on every push against a fresh database.
- UI changes are verified with real mouse-event browser testing before commit; production checks clean up their own test data.

6. Deployment & operations

ITEM	VALUE
Live URL	https://teams-collaboration-tool-bwfb.vercel.app
App hosting	Vercel, region Tokyo (<code>hnd1</code>)
Database	Supabase hosted project, Tokyo (<code>ap-northeast-1</code>)
Email sender	<code>noreply@algotix.ai</code> via Resend (domain verified)
Cost	All free tiers: Vercel Hobby, Supabase Free, Resend Free

Routine operations

TASK	HOW
Deploy an update	Push to the GitHub main branch — Vercel builds and deploys automatically.
Change a setting / key	Edit Environment Variables in the Vercel dashboard, then redeploy.
Database changes	Written as migration files and applied with the Supabase CLI (<code>supabase db push</code>).
Backups	Weekly <code>supabase db dump</code> is recommended (free plan has no automatic point-in-time recovery).
Keep-alive	Supabase free projects pause after ~7 days without traffic — a free daily ping (<code>cron-job.org</code>) keeps it awake.

Free-tier limits to know

- **Supabase Free:** 500 MB database, 1 GB file storage, 5 GB bandwidth/month, pauses when idle for a week.
- **Vercel Hobby:** non-commercial license, 100 GB bandwidth/month. A paid plan is the correct step once the company uses it commercially at scale.
- **Resend Free:** 100 emails/day, 3,000/month — far above invitation volume.

Planned / not yet built

- Push notifications when the browser is fully closed (current notifications need an open tab).
- Message content search.
- Virus scanning of uploaded files.
- Message edit and delete.

Project size: 28 database migrations · 89 commits · 81 unit tests + full e2e suite · 6 themes · 5 detector families — running in production since August 2026.